St. Paul School District 625 - 2013-2015 Labor Contract - An Evidence-Based Design Perspective

Thomas J. Smith (UM College of Education & Human Development School of Kinesiology)

To What Degree are Changes to the Union Contract Approved During the Recently Concluded St. Paul School District 625 Labor Dispute Supported by Findings from Evidence-Based Design Research Regarding the Efficacy of the These Changes in Improving Student Academic Achievement?

Key Conclusions: (1) On balance, terms of the 2013-2015 School District 625 settlement with the teachers union are aligned with findings from evidence-based design (EBD) research; (2) Yet it is disappointing that these findings were not cited (at least publically) in negotiating these terms; (3) With any effort to address student achievement, including contract negotiations, K-12 educators and administrators should emphasize allocation of resources to design factors for which a strongly positive impact on student achievement has been reliably demonstrated through EBD research. Investment in factors with equivocal, varied or weak influence on student achievement should be viewed with more caution and/or skepticism.

Introduction

► In late February, the Teachers Union and St. Paul School District 625 approved a new contract containing a number of changes to the previous contract.

► A primary goal of the new contract is to improve academic performance of students enrolled in District schools.

► The new contract mandates a series of changes in the design of learning environments, or in the design of the instructional system, in district schools.

► The analysis offered here addresses the question of whether the goal of these design changes --- improving student academic achievement --- is supported by findings from evidence-based design research regarding the efficacy of different educational system design interventions in improving student performance.

Background

► Evidence-based design (EBD) is defined as that field of study that emphasizes the application of credible evidence to influence design [1].

► EBD originally was popularized in health care as an approach to improve patient and staff well-being, patient healing, stress reduction, and patient safety [2].

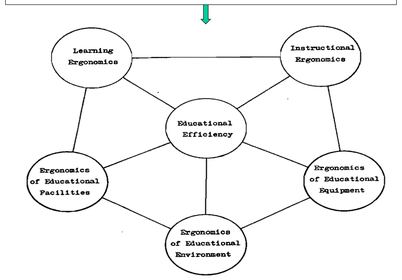
► The field of education largely has yet to recognize the value of EBD for making informed decisions about designing educational systems and environments to improve student performance.

► However, application of ***educational ergonomics*** can be used to support EBD research, practice, and design decision-making in the field of education.

Educational Ergonomics

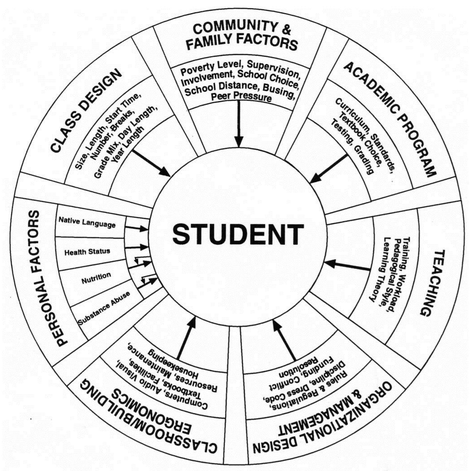
► Educational ergonomics is defined as that branch of E/HF concerned with the study of how and why student learning is influenced by the design of learning environments [3-8].

► EBD research, inspired by educational ergonomics, has identified a series of educational system and environment design factors that reliably benefit K-12 student academic performance.



The educational ergonomics model of Kao [4]

The educational ergonomics model of Smith [5]



Selected Design Factors Reliably Shown to Have a Strong Influence On Student Academic Achievement [6,8]

► Cooperative Learning

► Early Childhood Education

► Environmental Design of School Classrooms

► Longer School Days and/or School Year

► Later school day start times (for high school students)

► Physical Activity

► Community Socioeconomic Status

► Teaching Quality

Key Features of the St. Paul School District 625 2013-2015 Settlement with the Teachers Union [9,10]

► Reduced class sizes for kindergarten through high school classes. For kindergarten through third grade classes, somewhat smaller classes for higher- relative to lower-poverty classes

► Increased spending on early childhood education

► Increased teacher salaries

► More support staff (counselors, nurses, social workers, librarians)

► Community outreach emphasis through more parental involvement and home visit program

► Less emphasis on preparing students for taking standardized tests

The Merits of These Contract Features for Improving Student Academic Achievement are Assessed Below, using Letter Grades, Based on EBD Perspectives

Reduced Class Sizes

Grade: B

EBD Interpretation

Small classes benefit achievement of elementary (especially disadvantaged) students, but documented benefits for middle and high school students are inconclusive [11,12].

Increased Spending on Early Childhood Education

Grade: A

EBD Interpretation

Few design factors have been shown to exert a greater influence on the school performance, and on the lifelong success generally, of students than early childhood education [13].

Increased Teacher Salaries

Grade: C

EBD Interpretation

Higher salaries may improve the wellbeing and sense of self-worth of teachers, but there is no evidence regarding a link between teacher salary levels and student academic achievement [14].

More Support Staff (Counselors, Nurses, Social Workers, Librarians)

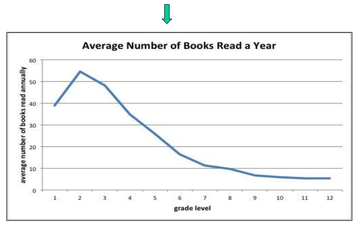
Grade: A-

EBD Interpretation

► Counselors, nurses & social workers can aid tracking of physical and mental health, home abuse/violence, nutritional status, physical activity, and substance abuse among students, all of which influence student academic achievement [15].

► However, reading among children shows a steady decline with grade level (see chart below) [16].

► Therefore, support for Librarian staffing should be assessed in relation to the degree to which improvements in student reading may be realized.



Community Outreach

Grade: A

EBD Interpretation

► The new contract calls for a rework of the parent-teacher format [10], and continued support for the teacher home visit program [17].

► Additional parental involvement in District affairs also is likely [18].

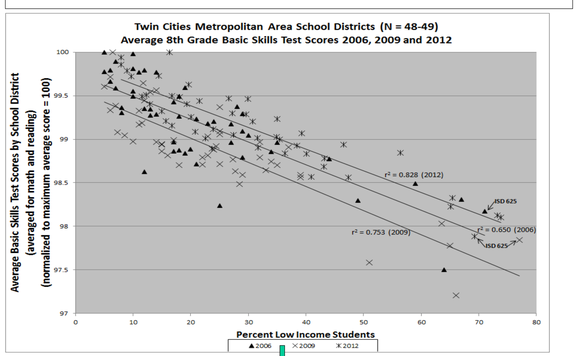
► There are intimate links between the socioeconomic status of a community and the academic performance of students attending schools in that community. [19].

► This conclusion is based, in part, on an analysis of average combined math and reading scores on Minnesota standardized tests by 8th graders in 48-50 Twin Cities metro-area school districts, for the years 1996, 2002, 2006, 2009 and 2012, in relation to the poverty index for these districts [19; 20 (Chap. 4)].

► The chart below shows results for the years 2006, 2009, 2012. For these 3 years, plus 1996 and 2002, r^2 values range from 0.65 to 0.83. That is, across these districts and these years, the poverty index accounts from about two-thirds to over four-fifths of the variance in average combined math and reading scores for 8th grade students.

► These results suggest that the ergonomic design of a community---its socioeconomics---is a major predictor of the academic performance of students attending schools in that community. For a school district to ignore the impact of the design of the surrounding community on student achievement thus represents a recipe for student failure.

► From this perspective, the innovative efforts of the St. Paul District to align its operations more closely with St. Paul parents, households and communities may pay dividends in terms of student academic achievement.



Less Emphasis on Preparing Students for Taking Standardized Tests

Grade: B

EBD Interpretation

► With the advent of standardized math and reading tests in Minnesota in 1996, preparing students for taking these tests has come to demand a substantial amount of time and effort in Minnesota schools.

► From this perspective, this contract feature is reasonable.

► However, standardized testing results allow comparisons of student performance across different school districts (see chart above).

Less Emphasis on Preparing Students for Taking Standardized Tests (continued)

► Across close to 50 Twin Cities metro area school districts, since 1996 District 625 8th grade students consistently have received among the lowest scores on standardized math and reading tests [19; 20 (Chap. 4)]. Standardized test results for these students for '06, '09, and '12 are flagged on the chart.

► Yet teachers have essentially no control over the socioeconomic conditions of the surrounding community.

► Standardized testing is not going away. To deal most effectively with this reality, the teachers union should mount a strenuous effort to convince all stakeholders that improving the socioeconomic design of urban communities (through community ergonomics) offers the best hope of improving academic performance of students in urban schools.

Conclusions

► On balance, terms of the 2013-2015 School District 625 settlement with the teachers union are aligned with findings from EBD research.

► Yet it is disappointing that these findings were not cited (at least publically) in negotiating these terms.

► With any effort to address student achievement, including contract negotiations, K-12 educators and administrators should emphasize allocation of resources to design factors for which a strongly positive impact on student achievement has been reliably demonstrated through EBD research. Investment in factors with equivocal, varied or weak influence on student achievement should be viewed with more caution and/or skepticism.

References

1. Rubin, A. (2008). *Practitioner's guide to using research for evidence based practice*. Hoboken, NJ: Wiley.
2. Ulrich, R.S., Zimring, C. Zhu, X., Dubose, J., Seo, H.-B., Choi, Y.-S., Quan, X., & Joseph, A. (2008). *A review of the research literature on evidence-based healthcare design*. Concord, CA: Center for Health Design.
3. Smith, K.U., & Smith, M.F. (1966). *Cybernetic principles of learning and educational design*. New York: Holt, Rinehart and Winston.
4. Kao, H.S.R. (1976). On educational ergonomics. *Ergonomics, 19*(6), 667-681.
5. Smith, T.J. (2007). The ergonomics of learning: educational design and learning performance. *Ergonomics, 50*(10), 1530-1546.
6. Smith, T.J. (2011). Ergonomics of learning environments – designs with strong, equivocal or poor returns on educational investment. In *Proceedings of the Human Factors and Ergonomics Society 55th Annual Meeting* (pp. 560-564). Santa Monica, CA: Human Factors and Ergonomics Society.
7. Smith, T.J. (2012). Integrating community ergonomics with educational ergonomics – designing community systems to support classroom learning. *Work: A Journal of Prevention, Assessment & Rehabilitation, 41*, 3676-3684.
8. Smith, T.J. (2013). Designing learning environments to promote student learning – ergonomics in all but name. *Work: A Journal of Prevention, Assessment & Rehabilitation, 44*, S39-S60.
9. http://www.mn2020hindsight.org/view/the-new-saint-paul-teacher-contract-whats-in-it
10. Koumpilova, M. (2014). Teacher deal goes beyond wages and class size. In *St. Paul Pioneer Press* (March 2), pp. 1A,5A.
11. Greene, J.P., & Winters, M.A. (2005). Five myths crying out for debunking. In: *National Review* (October 24), pp. 49-51.
12. Biddle, B.J., & Berliner, D.C. (2002). *What research says about small classes and their effects*. San Francisco, CA: WestEd. (www.WestEd.org/policyperspectives)
13. Barnett, W.S. (2011). Effectiveness of early educational intervention. *Science, 333*, 975-978.
14. Klein, J. (2011). Scenes from the class struggle. In: *The Atlantic* (June), pp. 66-77.
15. Swingle, C.A. (1997). *The relationship between the health of school age children and learning: implications for schools*. Lansing, MI: Michigan Department of Community Health.
16. Schutte, A. (2012). *So, how much are kids really reading?* (April 16). Yalsa the Hub (Your Connection to Teen Reads) (http://www.yalsa.ala.org/thehub/2012/04/16/so-how-much-are-kids-really-reading)
17. Lonetree, A. (2013). Making house calls, teachers build ties. In *St. Paul Pioneer Press* (August 25), pp. B1,B4.
18. Koumpilova, M. (2014). In St. Paul, parents eager to flex their muscle again. In *St. Paul Pioneer Press* (March 4), pp. 1A,2A.
19. Smith, T.J. (2012). Integrating community ergonomics with educational ergonomics – designing community systems to support classroom learning. *Work: A Journal of Prevention, Assessment & Rehabilitation, 41*, 3676-3684.
20. Smith, T.J., Henning, R., Wade, M.G., & Fisher, T. (2014). *Variability in human performance*. Boca Raton, FL: CRC Press. To be published.

CEHD Research Day, March 25, 2014